

Lightweight zero-knowledge authentication scheme for IoT embedded devices

Zhangquan Wang^a, Jiaxuan Huang^b, Kelei Miao^a, Xiaowen Lv^a, Yourong Chen^{a,*}, Bing Su^b, Liyuan Liu^c, Meng Han^{d,e}

^a College of Information Science and Technology, Zhejiang Shuren University, Hangzhou, 310015, China

^b School of Computer and Artificial Intelligence, Changzhou University, Changzhou, 213164, China

^c Department of Decision & System Sciences, Saint Joseph's University, Philadelphia, 19131, USA

^d Binjiang Institute of Zhejiang University, Hangzhou, 310053, China

^e Zhejiang University, Hangzhou, 310058, China

ARTICLE INFO

Keywords:

Zero-knowledge proof
Authentication
Lightweight
Embedded device
Internet of things

ABSTRACT

Authentication technology has been employed in a variety of Internet of Things (IoT) applications, such as electronic medical services, smart homes, and the Internet of Vehicles, to protect data privacy and device accessibility. However, current zero-knowledge authentication schemes for resource-limited embedded devices face challenges in computation, communication, and storage costs. In order to solve this problem, we propose a lightweight zero-knowledge authentication scheme for IoT-embedded devices (LZIA). In the proposed scheme, the identity information is associated with a unique identifier, and the device identities are allocated using hash commitment in the registration phase. Then, a lightweight registration method is proposed to reduce the computation and storage costs. In the secret key management phase, the direct communication between the embedded device and the authentication server (AS) for device key management is built. By combining device identity information and the Chebyshev polynomial chaotic map, we simplify the device's secret key structure and propose a secret key generation equation, an update equation, and a secret key management method to reduce computation and storage costs. In the authentication phase, we replace traditional cryptographic operations with Chebyshev polynomial operations and consolidate all intermediate data into a polynomial for unified authentication. We then optimize the device's proof generation and proof verification equations and propose a lightweight authentication protocol to effectively reduce the number of interactions and computational costs. Experimental results demonstrate that LZIA, whether in simulation or actual hardware environments, can reduce computation, communication, and storage costs while guaranteeing security. LZIA surpasses EEAS, LAKA, and SRAM in terms of both security and performance.

1. Introduction

In recent years, various IoT-based intelligent systems related to healthcare, home, transportation, etc. [1–3] are being developed rapidly. In the intelligent healthcare area, healthcare practitioners establish the connection between patients and medical institutions by utilizing intelligent medical devices to collect patient health data in real time. This data is integrated to identify the physical condition of patients and help to develop/adjust treatment plans more effectively [4]. As IoT technology popularized, it is necessary to pay attention to the security and privacy in IoT applications [5,6] where numerous security events have occurred. The IoT-embedded devices, which usually handle

sensitive data and are connected to larger networks, become particularly vulnerable. Without adequate authentication, these devices can be exploited maliciously, leading to data breaches or network intrusions. As the first crucial line of defence, authentication attracts lots of interest in IoT security and privacy [7]. With the limitations in storage, computation ability, and power consumption [8], traditional and intricate authentication schemes cannot support resource-constrained embedded devices. Therefore, it is critical to develop a lightweight and secure authentication scheme for resource-efficient devices.

The authentication schemes mainly include password authentication, biometric authentication, trip trajectory authentication, zero-knowledge proof, and so on. Password authentication schemes require

* Corresponding author.

E-mail addresses: 600389@zjsru.edu.cn (Z. Wang), s21150812059@smail.cczu.edu.cn (J. Huang), klmiao@zjsru.edu.cn (K. Miao), xwl1129@zjsru.edu.cn (X. Lv), chenyr@zjsru.edu.cn (Y. Chen), subing@cczu.edu.cn (B. Su), lliu@zju.edu.cn (L. Liu), mhan@zju.edu.cn (M. Han).

<https://doi.org/10.1016/j.comnet.2023.110021>

Received 21 June 2023; Received in revised form 18 August 2023; Accepted 5 September 2023

Available online 16 September 2023

1389-1286/© 2023 Elsevier B.V. All rights reserved.

a username and password from user input, which is subsequently sent to the device in ciphertext to authenticate their identity. This approach entails storing sensitive user data on the device and is susceptible to attacks, leading to the risk of potential key and user information leaking [9]. Biometric authentication schemes utilize user features, such as fingerprints and facial characteristics, to authenticate their identity. However, it usually applies artificial intelligence algorithms to achieve high accuracy in classifying feature data, resulting in high computing costs and not suitable for IoT devices with low computing power [10]. Trip trajectory authentication schemes use personal trip trajectories and stay points to complete authentication. However, it proves challenging to authenticate users who frequently alter their trip trajectories and stay points, thus leading to low availability [11]. Zero-knowledge proof is a method that enables a prover to persuade a verifier of the veracity of a statement without disclosing any valuable information to the verifier. It allows the prover to prove their identity to the verifier and its legitimacy confidently without disclosing any private information linked to their identity [12]. Therefore, zero-knowledge proof is a proper scheme for IoT devices with limited resources, since it does not need high computing power. More importantly, it can achieve secure authentication and avoid privacy disclosure simultaneously during the authentication. Considering the characteristics of enhanced security, privacy protection, and lightweight processing, zero-knowledge proof authentication acts as a particularly suitable and practical scheme for IoT-embedded device authentication.

However, embedded devices that possess limited computational and storage capacity may allocate a significant amount of resources toward application services, thereby resulting in limited resources for security protection. Applying zero-knowledge authentication schemes in IoT-embedded devices is challenging in the following three aspects:

- (1) The registration process requires computing and storing large-volume data, which cannot meet the lightweight requirements of embedded devices.
- (2) The device key with a complex structure requires high computation and storage costs. Moreover, the existence of a trusted center for managing the secret key increases the communication cost in the secret key distribution phase and introduces issues in network delay and center trust security.
- (3) Zero-knowledge proof methods require complex interactive authentication procedures with high computational, communication, and storage expenses which prolong authentication time and lose the possibility of real-time.

Considering the above problems, we propose a lightweight zero-knowledge authentication scheme for the embedded devices of the Internet of Things (LZIA). The contributions of this paper are as follows:

1. To reduce computation and storage costs in the registration phase, we propose an innovative lightweight registration method where the device *ID* is bound with its random number, which is used to verify the device's validity. Then, the characteristics of hash commitments, which characterize both the hiding and binding properties, are used to allocate device identities. As such, the computation and storage costs are reduced greatly.

2. To reduce the communication and computation costs in the secret key distribution phase, we propose an innovative secret key management method. This method simplifies the device key structure by using direct communication between the device and the authentication server. Then we propose a key generation equation based on only two integers by combining the device identity information and Chebyshev polynomial chaotic map and an update equation to eliminate the need for recalculating public and private keys. Furthermore, a secret key management method is proposed to provide valid secret keys for the generation and verification of proof.

3. To effectively reduce the number of interactions and computation costs in the authentication phase, we propose a lightweight authentication protocol based on interactive zero-knowledge proof.

The Chebyshev polynomial is used to reduce the computational complexity to replace the traditional complex cryptographic operations such as modular exponentiation, elliptic curve cryptography, etc. By integrating all intermediate data into a polynomial, the verification process is unified. Furthermore, the proof generation equation and the proof verification equation are improved based on the three challenge values. Finally, we propose the lightweight interactive zero-knowledge authentication process to minimize the number of interactions.

The remaining content is organized as follows. Section 2 presents the current authentication schemes. Section 3 introduces some general concepts. Section 4 describes the principles and processes of the proposed scheme. Section 5 analyzes the security of the proposed scheme. Section 6 conducts experimental simulations to compare and analyze schemes, and Section 7 draws conclusions.

2. Related work

Authentication is widely applied in various fields, such as intelligent transportation, intelligent healthcare, and intelligent home. Some researchers focus on password-based authentication schemes by checking whether users provide correct passwords. Upon successful validation of the password, the system issues an identity verification token to the user, which can be utilized to access data, request services, or apply for other permissions. For example, Agrawal et al. [13] introduced the concept of password-based threshold token authentication. It retrieved the server-derived password in an unrelated manner and calculated the corresponding credentials for identity verification. It could ensure the security of both password and identity verification tokens, but it was unfortunately vulnerable to password-guessing attacks. This scheme was improved by Zhang et al. [14], who proposed a threshold single sign-on authentication scheme. The scheme used multiple identity servers to authenticate mobile users and distributed authentication tokens in a threshold manner. It allows identity servers to update the secret shares regularly to effectively defend against key guessing attacks. However, this method is needed to store the encrypted information of authentication data on devices and multiple servers, facing problems such as password forgetfulness, loss, and leakage [13,14], and they are difficult to be applied to an environment where the identity servers are untrustworthy. Since each person has unique biological features, users no longer need to remember their passwords if biological features can be used as passwords. Therefore, some scholars focus on authentication schemes based on biometric recognition. For example, Zhang et al. [15] proposed a multimodal biometric recognition authentication system based on both electroencephalogram (EEG) and gait signals. This system used an invalid ID filtering model to intercept unauthorized users. It also used an identification model based on an attention-based recurrent neural network for parallel recognition of the user's EEG and gait signals. However, it usually relied on artificial intelligence and other technologies to classify feature data accurately, where the high computation cost is unacceptable for IoT devices with low computing power. At the same time, some scholars focus on authentication schemes based on trajectory recognition. Gao et al. [16] proposed a trajectory-based authentication scheme which could judge whether the user is valid by calculating the similarity between the sample and template trajectories. However, it is hard to verify the users whose travel trajectories are changed frequently.

The zero-knowledge proof method can guarantee that the verifier remains uninformative about the verifier's identity key during the whole authentication phase. Compared with other methods, this method can significantly improve the security of the authentication phase. Therefore, several scholars have applied zero-knowledge proof methods to authentication, as shown in Table 1. Zero-knowledge authentication schemes can be categorized into two types: secure authentication schemes and lightweight authentication schemes. Secure authentication schemes primarily concentrate on defending against

Table 1

Summary of authentication schemes based on zero-knowledge proof.

Literature	Category	Purpose	Security	Methods	Disadvantages
[17]	Security authentication	Minimizes potential for successful deception by adversaries	Relies on MPP	Equation derivation, Hash	Complex protocol construction, high computation cost
[18]		Implements secure authentication in RFID systems	Relies on QRP	Equation derivation, Hash	
[19]		Ensures the security of data transmission	Relies on DLP	Sigma protocol, Rijndael algorithm	
[20]		Implements distributed authentication		Bilinear mapping, Ring signature algorithm	
[21]		Provides privacy and anonymity		Improved Schnorr protocol, ZKNimble algorithm	
[22]	Light-weight authentication	Non-IoT devices focused	Relies on DLP	Equation derivation, DH algorithm	Dependent on trusted third-party for key management
[23]		Implements user identity privacy protection scheme		SE, Modular exponentiation	
[24]		Implements lightweight authentication against guess attacks	Relies on QRP	Improved FFS protocol	
[25]		IoT devices focused	Relies on DLP	FO commitment, EEC	Complex registration and secret key structure, high computation and storage costs due to various operations (e.g., ECC and modular exponentiation)
[26]				Improved Schnorr protocol, ECC, SE	
[27]				ECDH algorithm, AE	
[28]				Hash, XOR, Modular exponentiation	
[29]				ECC, MAC, Hash, XOR, Modular exponentiation	
[30]				Chebyshev polynomial, Hash, XOR, Modular exponentiation	
[31]				Chebyshev polynomial, Hash, Modular exponentiation	
[32]				Chebyshev polynomial, Hash, XOR	

quantum computing attacks, and their security is dependent on different NP-hard problems. For example, Wang et al. [17] proposed a five-round zero-knowledge authentication scheme based on the matrix padding problem (MPP). It improved the three-round authentication scheme by adding a random challenge value communication between the verifier and the prover. This scheme could reduce the probability of deception in a single round of authentication from $2/3$ to $1/2$. But the scheme required many repetitions of authentication to achieve the expected security. Song et al. [18] proposed a zero-knowledge authentication scheme based on the quadratic residue problem (QRP). This scheme used a random oracle to establish an indistinguishable privacy model. It has been demonstrated that the complexity of QRP calculation and the hash function can obtain robust privacy. Moreover, the scheme is implemented for the identity verification of RFID tags. Soewito et al. [19] proposed an IoT security system. More specially, it improved the Sigma protocol to achieve authentication. Then, it

used the Rijndael algorithm to encrypt the data and send ciphertext to the server. It can enable secure data interaction, but no analysis of authentication scheme security was presented. Andola et al. [20] used non-interactive zero-knowledge proof of bilinear mapping and ring signature algorithms to implement anonymous authentication of drones on the blockchain, but this scheme involves complex algorithms such as bilinear mapping and ring signatures, and the computation is relatively large. Ashutosh et al. [21] computed the commitment hash value by the prover as the challenge value for the verifier, and enhanced the efficiency of the traditional interactive Schnorr protocol. Then, they further integrated the ZKNimble encryption algorithm to validate device identities. However, the protocol involves modular exponentiation, which requires considerable computational resources and is not suitable for resource-limited devices. The methods in [17–21] effectively ensured the security of secret data such as private keys. However, these protocols' constructions were complex. They had

high computation costs for authentication, and were unavailable for resource-constrained IoT devices.

Therefore, some scholars focus on lightweight zero-knowledge authentication schemes. For example, Feng et al. [22] proposed a lightweight collaborative authentication with key protection for a smart electronic health record (EHR) system. In the setup phase, the secret key generation center generated system parameters and distributed keys. This scheme used modular exponentiation, hash, XOR, and other operations in the collaborative authentication phase to verify identity. Then, it established a public session key using the Diffie–Hellman (DH) algorithm for further secure communication. Ren et al. [23] proposed a lightweight scheme for the protection of user identity privacy. The scheme introduced a trusted third party to manage user information and used an interactive zero-knowledge proof protocol to verify user identity based on the discrete logarithm problem (DLP). Then, it established a secure channel for the user to ensure regular access to data through symmetric encryption (SE). Han et al. [24] proposed an efficient and secure authentication scheme based on the Feige–Fiat–Shamir (FFS) protocol. The scheme used the method of zero-one reversal and two-to-one verification to defend against guessing attacks. The [22–24] achieved user identity privacy protection and common attack defence. However, these studies required a trusted center to manage identity information, such as keys. This approach renders the risk of widespread data privacy breaches and is difficult to apply directly to IoT devices.

Consequently, some scholars have shifted their focus toward developing secure and lightweight zero-knowledge authentication schemes tailored to IoT applications. For example, Xi et al. [25] used lightweight Fujisaki–Okamoto (FO) commitment and elliptic curve cryptography (ECC) to achieve secure and efficient authentication. However, the scheme needed to maintain a revocation list, and had to check whether the current user was on the revocation list during each verification process. It reduced authentication efficiency and increased additional storage costs. Antonio et al. [26] used a secure hash function to generate random challenge values of the verifier. This method effectively eliminates the necessity for multiple interactions during the authentication process, thereby reducing the device's costs. However, despite these efficiencies, the scheme involves the use of ECC and SE, which ultimately results in high computation costs. Verma et al. [27] proposed a lightweight and secure authentication protocol. It used the elliptic curve Diffie–Hellman (ECDH) algorithm, asymmetric encryption (AE), hash function, and timestamps to ensure authentication security. However, the cost of ECDH is relatively high. Jain et al. [28] proposed a simple and lightweight mutual authentication protocol for WSNs. This protocol stored the identity of each node in other nodes and used simple one-way hash functions together with XOR operations to achieve security levels. This scheme's computation cost was low, but it was not suitable for low-power sensor devices. Kumar et al. [29] proposed a lightweight authentication and key agreement protocol (LAKA). This protocol used the hash function and ECC to construct keys in the registration phase. Then, it used ECC, hash function, and message authentication code (MAC) to achieve two-way authentication. However, the introduction of these operations (ECC, modular exponentiation, etc.) would increase the computation cost of registration and authentication. Considering that the Chebyshev chaotic map has a lower computation cost, Liu et al. [30] proposed a multi-factor authentication scheme (SRAM). Each user had multiple authentication factors. These factors could be remotely authenticated by the server or gateway node and participate in the secret key protocol phase. This scheme used Chebyshev chaotic map to calculate multiple authentication factors as proof to verify their validity. Boubakri et al. [31] used the arithmetic relationship of the semi-group property based on Chebyshev chaotic mapping, the order of the polynomial, and the output of the polynomial itself, and constructed a zero-knowledge proof protocol for cyber-physical system (CPS). This protocol is lightweight enough in a single

round of verification. However, the data interacted in each authentication process is not random, so it requires multiple verifications to resist theft attacks. Zhang et al. [32] proposed an authentication scheme (EEAS). This scheme used a binary power algorithm based on matrices to provide secure and efficient Chebyshev polynomial calculations. Then it combined lightweight cryptographic elements such as hash and XOR to construct a fast authentication and key negotiation scheme. This scheme reduced the computation and communication costs in the authentication phase. The methods in [25–32] enhance verification security and efficiency through technologies such as Chebyshev polynomials and ECC. However, these studies are complex in the registration method and secret key structure. Furthermore, the authentication phase in these schemes still requires performing numerous operations, including modular exponentiation, ECC, and hash, to calculate and verify intermediate data used for authentication. This can still result in high computation and storage costs.

3. Background

This section aims to provide a brief introduction to fundamental concepts and definitions utilized in this paper.

3.1. Zero-knowledge proof

Definition 1 (Zero-knowledge proof). The Zero-knowledge proof is an agreement involving two parties. It requires the prover P to prove to the verifier V that he knows a secret, but the verifier V does not obtain any useful information from prover P [12]. Therefore, the zero-knowledge proof has been widely used in authentication. More specifically, the prover P does not want to disclose the private key and realizes authentication by submitting proof to the verifier V . No one can obtain any knowledge related to identity information from the proof. The zero-knowledge proof protocol is considered to be safe if it owns the following three characteristics.

- (1) Completeness: If the prover's statement is true, the verifier can always believe that the prover owns the secret.
- (2) Soundness: If the prover's statement is false, It is almost impossible for the prover to authenticate successfully. In other words, the probability that the adversaries successfully deceive the verifier by providing a false identity certificate can be ignored.
- (3) Zero-knowledge: If the prover has the identity information (e.g., the private key), the verifier cannot derive any valid information related to the identity information from the proof.

3.2. Chebyshev polynomial chaotic maps

Definition 2 (Chebyshev Polynomial). The Chebyshev polynomials are a series of orthogonal polynomial sequences in recursive ways [30] and are defined as:

$$T_n(x) = \begin{cases} 1, & n=0 \\ x, & n=1 \\ (2 \times x \times T_{n-1}(x) - T_{n-2}(x)) \bmod p, & n \geq 2 \end{cases} \quad (1)$$

where x represents the variable parameter, $T_n(x)$ represents the n -degree polynomial with parameter x , n represents an integer, and p represents large prime number. Chebyshev polynomials have the following two essential properties:

- (1) Semigroup property, which is shown as:

$$T_n(T_r(x)) \bmod p = T_r(T_n(x)) \bmod p = T_{n \times r}(x) \bmod p \quad (2)$$

- (2) The arithmetic relationship between the polynomial degree and the output of the polynomial itself can be characterized as:

$$2 \times T_n(x) \times T_r(x) \bmod p = T_{n+r}(x) \bmod p + T_{|n-r|}(x) \bmod p \quad (3)$$

Definition 3 (Chaotic Map-Based Discrete Logarithm Problem). Given x and y , it is almost impossible to find the integer n , such that $T_n(x) = y$.

Definition 4 (Chaotic Map-Based Diffie-Hellman Problem). Given x , $T_r(x)$ and $T_n(x)$, it is almost impossible to find $T_{r \times n}(x)$.

3.3. Common types of attacks

Referring to the anti-attack security requirements of authentication in sensor networks [29,32], we introduce seven classic attacks against the authentication scheme:

Definition 5 (Replay Attack). The adversary eavesdrops on the authentication data in the network and then repeatedly sends it to the verifier to obtain illegal services or benefits.

Definition 6 (Counterfeit Attack). The adversary intercepts the identity information, such as the public key of the legitimate device in the network, and uses the identity information to fake the legitimate device to implement deception.

Definition 7 (Sensor Node Capture Attack). Assume that the adversary has sufficient computing power and means to eavesdrop and obtain the private information of some legitimate devices in the network. Thus, the adversary can attack more communication links in the network.

Definition 8 (Man-in-the-middle Attack). The adversary intercepts the communication data between the device and the AS through technical means. So the adversary can perform data tampering, sniffing, and other operations without the awareness of both parties.

Definition 9 (Server Insider Attack). Assume that the adversary is an insider of the AS. The adversary knows the registration information of all IoT devices and can use the known registration information to obtain illegal services or benefits.

Definition 10 (Privileged Insider Attack). Assume that the adversary is an internal developer of the IoT device. The adversary can freely obtain the identity information stored in the device and can use known identity information to get illegal services or benefits.

Definition 11 (Guessing Attack). The adversary can continuously try different numbers to guess the private data of the IoT device until access is obtained.

4. Proposed authentication scheme

Some important notations employed in this paper are described in Table 2.

As shown in Fig. 1, the IoT system consists of embedded devices, the cloud, and the AS. The embedded devices communicate with the AS via a cloud network. The AS is responsible for registering, verifying identities, and updating keys for embedded devices. Embedded devices that successfully complete authentication are permitted to access the network and undertake real-time information monitoring, remote control, and other functions. Nonetheless, their computing power and storage capacity are constrained, rendering them incapable of achieving more intricate device authentication.

Therefore, we propose a lightweight zero-knowledge authentication scheme for IoT-embedded devices based on the above architecture model, as shown in Fig. 2. Firstly, we improve the registration method. Specifically, we retrieve the random number of the embedded device to be registered and refine the encryption technique applied to the random number. The AS generates a unique identifier ID based on the encrypted random number and stores it in the local memory of the AS. Secondly, we improve the secret key management method. To be

Table 2

Notations and definitions.

Notation	Definition
ID	Unique identifier of the embedded device
TID	Unique identifier of the AS
x	The seed of Chebyshev polynomial
sk	Private key of the embedded device
$pk(x)$	Public key of the embedded device
p	Large prime number
$H(\cdot)$	Secure hash function
C	Commitment value
Z_p	Finite field
$ $	Concatenation operation

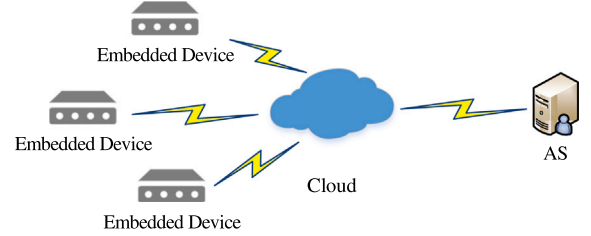


Fig. 1. IoT architecture model.

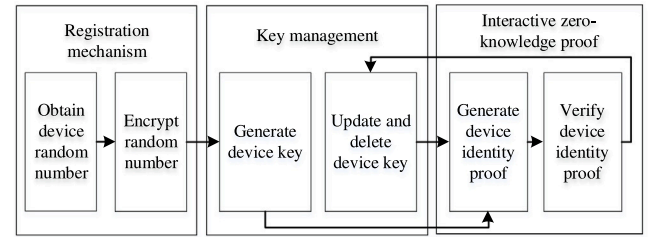


Fig. 2. Schematic diagram of LZIA.

precise, we enhance the generation and updating of embedded device keys by leveraging the unique identifier ID . In the next authentication, the embedded device keys are no longer to be re-registered and re-generated again. Moreover, the AS will remove device keys that have remained inactive for an extended period. Finally, we introduce a novel interactive zero-knowledge authentication protocol. Namely, it improves the generation method of the device's proof. This protocol facilitates the generation of the device's proof based on the random challenge value supplied by the AS and enhances the device's proof verification method. This protocol combines Chebyshev polynomials to verify the validity of proof and identity information. Then, the verification results are fed back to the device for the authentication of the embedded device. The details are as follows:

4.1. Registration method

Upon introducing a new embedded device to the IoT system, the AS must initiate the device to enable authentication. The resources of embedded devices are limited because of their lower computing power and storage capacity compared with other devices. However, the mainstream registration methods consume significant computation and storage resources, making them impossible in these lightweight embedded devices. Therefore, to reduce the amount of information required in the registration phase, we bind the device ID with its random number and verify the validity of the device's random number by checking the device ID informatively. Consequently, our proposed lightweight registration method allocates identities for embedded devices and reduces computation and storage costs in the registration phase.

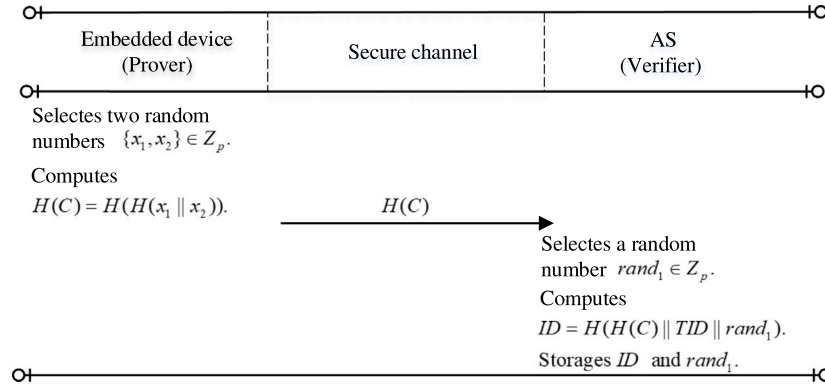


Fig. 3. Process diagram of the registration method.

In the traditional registration phase, each device usually has to store multiple pieces of information such as identifier ID , login password and secret random numbers. The AS also needs to store the ciphertext corresponding to each device information and its own identifier TID , secret key and other information. Each device's multiple pieces of information are distinct from each other. Thus, the AS must perform more XOR, hash, and other operations. The AS is obliged to calculate and verify all the device information one by one to check whether the information provided by the device is valid. It leads to high computation and storage costs in embedded devices. To decrease the costs incurred by embedded devices during the registration phase, we take advantage of the characteristics of both hiding and binding in hash commitment. Hash commitment is capable of not only concealing the information itself but also swiftly validating the relationship between the information. Therefore, we improve the registration method as shown in Fig. 3. Note that only the random number is considered, and the encryption method for the random number is improved. The specific process is as follows:

Step 1: The embedded device generates two random numbers from the finite field Z_p , which are used as a login password x_1 and a secret random number x_2 . Then, it calculates the commitment value $C = H(x_1 \| x_2)$ and its hash value $H(C)$. After that, the embedded device takes the hash value $H(C)$ as the random numbers' ciphertext and then sends it to the AS to reduce the computation cost of device encryption.

Step 2: The AS generates a random number $rand_1 \in Z_p$. Then, the AS calculates and stores the current embedded device's unique identifier $ID = H(H(C) \| TID \| rand_1)$ according to identifier TID and hash value $H(C)$. After that, the AS sends it to the embedded device to achieve the binding of the device ID and identity information $\{x_1, x_2\}$.

Step 3: The AS can initially verify the device by checking the ID , and allow it to perform authentication, thus reducing the storage cost of the AS.

4.2. Key management

After completing the above registration of IoT devices, we improve the secret key of each registered device. The device is subsequently able to utilize this key for authentication. Additionally, since embedded devices in the IoT system may frequently change, we propose a method for periodically updating the secret key of each embedded device to enhance the security of the authentication scheme and reduce the memory occupancy of invalid information. Furthermore, we propose to delete the secret key of a damaged device where details are as follows:

In terms of key generation: The mainstream key structures are relatively complex. Because they use structures including random sequence and matrix as private keys and generate public keys through modular exponentiation and other methods. These structures can result in large device key sizes, high computation cost and high storage cost. To reduce the storage and computation costs of the device key, we

simplify the secret key structure and use Chebyshev chaotic map on the finite field with less computation cost [29]. Therefore, we introduce Chebyshev chaotic map, improve the secret key structure and propose a novel key generation equation. Namely, we consider two integers as the private and public keys, respectively, instead of a random sequence and matrix with a complex structure. Then, we combine the device identifier ID , device number $\{x_1, x_2\}$, and randomly generated sequence $S = \{s_1, s_2\} \in Z_p$ to generate the newly registered device's public and private keys through Eq. (4). Thus, it can effectively reduce the computation and storage costs of the secret key.

$$\begin{cases} sk = ID \times s_1^{x_1} \times s_2^{x_2} \bmod p \\ pk(x) = T_{sk}(x) \bmod p \end{cases} \quad (4)$$

where sk represents the private key of the current newly registered device, and $pk(x)$ represents the public key of the current newly registered device, whose value can be freely changed according to the size of the variable x . To prevent adversaries from successfully calculating the private key, the sequence S needs to be erased from the device memory immediately after generating the private key.

In terms of key updates: Although the trusted key management center can reduce the device's computation cost to a certain extent, embedded devices exhibit characteristics of a significant quantity and frequent communication. As the number of devices in the network increases, the computation and communication costs of the secret key management center escalate. It is easy to cause traffic congestion and network delay problems, resulting in lower authentication efficiency and failure to meet the real-time requirements of the embedded devices [32]. Therefore, we improve the device key update method based on the secret key generation. Namely, we propose a novel key update equation so that the public and private keys do not need to be recalculated, resulting in the reduction of the computation cost.

$$\begin{cases} sk' = sk + rand_2 \\ pk'(x) = T_{sk'}(x) \bmod p \end{cases} \quad (5)$$

where sk' , $pk'(x)$, and $rand_2$ respectively represent the updated device private key, the updated device public key, and a random number with an interval of $(-p-1, p-1)$. We simplify the secret key update process, cancel the trusted key management center, and let the embedded device interact directly with the AS, thus can reduce the communication cost.

Then, we propose a secret key management method based on the above improvements, as shown in Fig. 4. LZIA outputs public and private keys (pk, sk) for each embedded device and can regularly update their key, thus providing valid secret keys for the generation and verification of proof. The specific process is as follows:

Step 1: If the current embedded device does not have secret keys, the device will generate a random sequence $\{s_1, s_2\} \in Z_p$. Then, it will calculate the private key sk and public key $pk(x)$ by Eq. (4).

Step 2: If the current embedded device has secret keys and the validity period is about to expire, the device will update the public

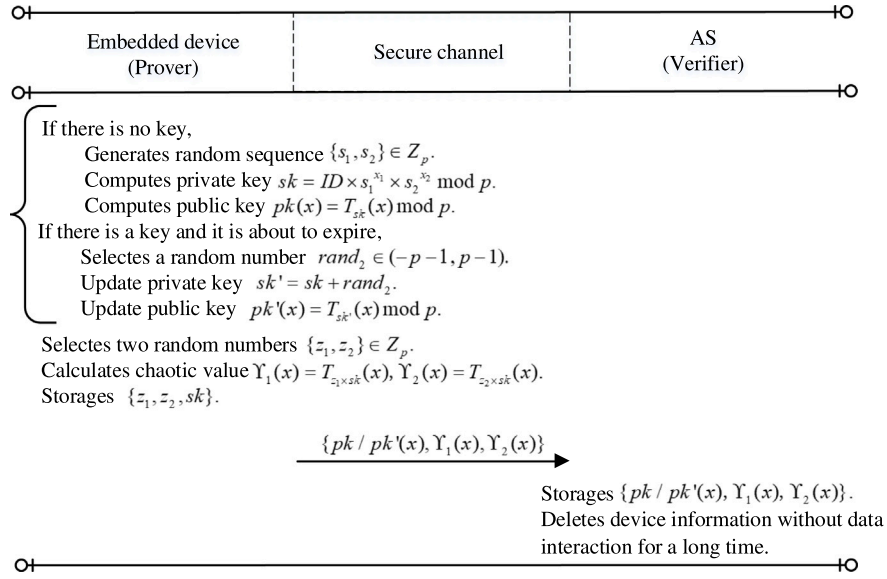


Fig. 4. Process diagram of secret key management method.

and private keys. Specifically, the device will randomly select a value $rand_2 \in (-p-1, p-1)$ and update the private key sk' together with the public key $pk'(x)$ by Eq. (5).

Step 3: If the embedded device completes the generation or update of the public and private keys, the device will select two random numbers $\{z_1, z_2\} \in Z_p$ as the proof key. Then, it will calculate the corresponding chaotic value $Y_1(x) = T_{z_1 \times sk}(x)$ and $Y_2(x) = T_{z_2 \times sk}(x)$, thus can prevent the adversary from deceiving the AS through guessing attacks. After that, the device will send chaotic values $\{Y_1(x), Y_2(x)\}$ and its public key to the AS. The AS will locally store chaotic values $\{Y_1(x), Y_2(x)\}$ and the public key of the current device.

Step 4: Suppose an embedded device fails, and the AS does not receive any data from the device for a long time. In that case, the AS will delete the device identifier ID and corresponding public key to reduce the memory space occupied by the damaged device information.

4.3. Interactive zero-knowledge authentication

Upon completing the aforementioned key generation or update of the embedded device, the device uses its private key to generate the proof. Then it combines the device ID and number $\{x_1, x_2\}$ to achieve lightweight authentication. On the one hand, the AS verifies the relationship between the embedded device and the secret key by checking the validity of the device information; On the other hand, the AS verifies the relationship between the public and private keys of the device by validating the proof. Only the embedded devices that pass all verifications can achieve authentication, thus achieving data transmission, remote control, and other functions between devices. However, due to the embedded devices' low computing and storage capacity, the resources invested in security protection are minimal. Moreover, the authentication process of the traditional zero-knowledge proof scheme is complex, and its computation, communication, and storage costs are elevated. The execution time of the traditional scheme is prolonged, and thus cannot satisfy the real-time requirements of embedded devices. Note that Chebyshev polynomial operation has a lower computation cost than the modular exponentiation operation. Thus, we replace the modular exponentiation operation of traditional zero-knowledge proof with the Chebyshev polynomial operation. Moreover, we present the generation and verification equation of the proof, along with a lightweight zero-knowledge authentication protocol, to facilitate the prompt authentication of embedded devices. The details are as follows:

Firstly, we propose an equation to generate the proof of the device. Specifically, we eliminate the modular exponentiation operation and employ distinct equations to generate the proof according to the designated challenge value. This way, we can eliminate the computation cost of the random challenge value on the proof and diminish the number of interactions. Additionally, we reduce the number of member variables used to compute the proof, thereby effectively reducing the computation cost of the proof.

$$\begin{cases} c_1 = sk \times z_1 \times rand_3 \times H(ID \| TS), q = 1 \\ c_2 = sk \times z_2 \times rand_3 \times H(ID \| TS), q = 2 \\ c_3 = sk \times rand_3 + H(ID \| TS), q = 3 \end{cases} \quad (6)$$

where c_1 represents the proof generated by the embedded device when the challenge value $q=1$; c_2 represents the proof generated by the embedded device when the challenge value $q=2$; c_3 represents the proof generated by the embedded device when the challenge value $q=3$; q represents the random challenge value generated by the AS, TS represents the current timestamp.

Secondly, according to the proof provided by the device, we propose an equation for verifying the device's proof. It can improve the authentication efficiency of the device while ensuring security and can reduce the computation and storage costs of verification. Specifically, the AS takes the generated hash value $H(ID \| TS)$, chaotic value $\{Y_1(Q), Y_2(Q)\}$ and other intermediate data as the parameters of the Chebyshev polynomial. Then, the AS adds them to the polynomial and generates the verification key vk_q through Equation (7).

$$\begin{cases} vk_1 = T_{H(ID \| TS)}(Y_1(Q)) \mod p, q = 1 \\ vk_2 = T_{H(ID \| TS)}(Y_2(Q)) \mod p, q = 2 \\ vk_3 = 2 \times pk(Q) \times T_{H(ID \| TS)}(q) \mod p, q = 3 \end{cases} \quad (7)$$

where vk_1, vk_2, vk_3 represent the verification keys under different challenge values, and Q represents a random chaotic value. Then, we take two essential properties of Chebyshev polynomials as the verification conditions of proof. Namely, we check whether the verification key vk_q , the proof c_q and the identity information C' provided by the embedded device meet Equation (8). If it is satisfied, the authentication of the embedded device is successful.

$$\begin{cases} H(H(C') \| TID \| rand_1) = TS \\ TS' - TS = \Delta TS, \\ vk_1 = T_{c_1}(q) \mod p, q = 1 \\ vk_2 = T_{c_2}(q) \mod p, q = 2 \\ vk_3 = T_{c_3 - 2 \times H(ID \| TS)}(q) + T_{c_3}(q) \mod p, q = 3 \end{cases} \quad (8)$$

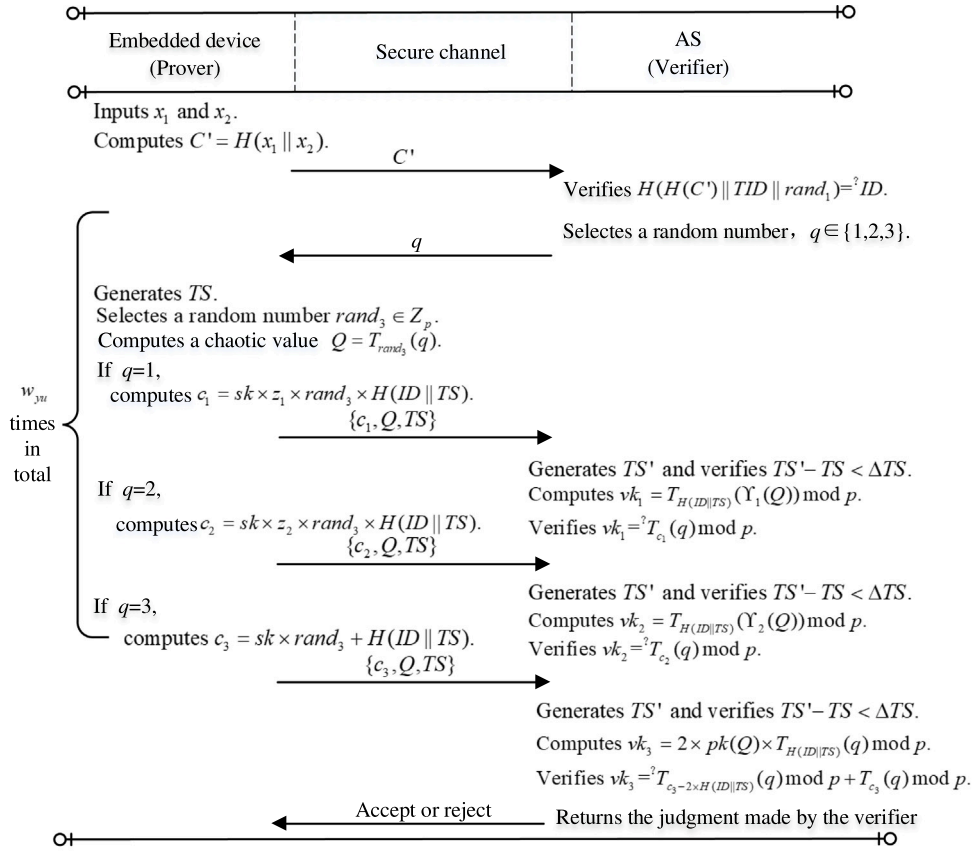


Fig. 5. Process diagram of zero-knowledge authentication.

where $=^?$ represents whether they are equal, TS' represents the current timestamp generated by AS; ΔTS is the average time threshold (experience value) for the AS to complete multiple authentications with the device, preventing messages from being replayed. We consider that malicious devices can fake normal devices to achieve illegal authentication by guessing the challenge value q and proof c_q . Therefore, the interactive protocol can repeat the generation and verification of the device's proof for w_{yu} times. The w_{yu} is calculated by dividing the expected size of the finite field by the actual size of the finite field. It represents the number of information exchange rounds between the prover and the verifier in the authentication protocol, which is used to enhance the credibility and security of the proof. The limited resources of embedded devices cannot handle excessively large finite field values. Therefore, we can reduce the actual size of the finite field and conduct w_{yu} rounds of verification consecutively to achieve the desired level of security.

Finally, as shown in Fig. 5, we propose a lightweight interactive zero-knowledge authentication protocol to authenticate the embedded device rapidly. The specific process is as follows:

Step 1: The embedded device calculates the commitment $C' = H(x_1 || x_2)$ based on the locally stored serial number and sends the commitment C' to the AS.

Step 2: The AS calculates $H(H(C') || TID || rand_1)$ and verifies whether it equals the device ID . If they are equal, let $w = 1$, and then the AS further verifies the device identity. Otherwise, the device authentication fails.

Step 3: The AS selects a random number $q \in \{1, 2, 3\}$ as the challenge value and sends q to the device. The embedded device generates timestamp TS and selects a random number $rand_3 \in Z_p$. Then, the device must calculate the chaotic value Q according to the challenge value q and a random number $rand_3$, and respond differently to the AS.

Step 4: If $q = 1$, the device calculates the proof $c_1 = sk \times z_1 \times rand_3 \times H(ID || TS)$, and sends c_1 , chaotic value Q and timestamp TS

to the AS. After receiving the proof, the AS generates the current timestamp TS' , and verifies that $TS' - TS$ is less than ΔTS . Then, the AS calculates the verification key $vk_1 = T_{H(ID||TS)}(Y_1(Q)) \bmod p$ based on the Chebyshev polynomial $Y_1(x)$ of the current device stored locally and verifies whether it is equal to $T_{c_1}(q) \bmod p$. If they are equal, the verification is successful when $q = 1$. Then, the AS records the verification result.

Step 5: If $q = 2$, the device calculates the proof $c_2 = sk \times z_2 \times rand_3 \times H(ID || TS)$, and sends c_2 , chaotic value Q and timestamp TS to the AS. After receiving the proof, the AS generates the current timestamp TS' , and verifies that $TS' - TS$ is less than ΔTS . Then, the AS calculates the verification key $vk_2 = T_{H(ID||TS)}(Y_2(Q)) \bmod p$ based on the Chebyshev polynomial $Y_2(x)$ of the current device stored locally, and verifies whether it is equal to $T_{c_2}(q) \bmod p$. If they are equal, the verification is successful when $q = 2$. Then, the AS records the verification result.

Step 6: If $q = 3$, the device calculates the proof $c_3 = sk \times rand_3 + H(ID || TS)$, and sends c_3 , chaotic value Q and timestamp TS to the AS. After receiving the proof, the AS generates the current timestamp TS' , and verifies that $TS' - TS$ is less than ΔTS . Then, the AS calculates the verification key $vk_3 = 2 \times pk(Q) \times T_{H(ID||TS)}(q) \bmod p$ based on the Chebyshev polynomial $pk(x)$ of the current device stored locally, and verifies whether it is equal to $T_{c_3-2 \times H(ID||TS)}(q) \bmod p + T_{c_3}(q) \bmod p$. If they are equal, the verification is successful when $q = 3$. Then, the AS records the verification result.

Step 7: Current verification times $w = w + 1$. If w is not greater than w_{yu} , skip to Step 3 again. Otherwise, the AS checks all verification results. If all verification results are successful under different q values, the AS accepts the zero-knowledge proof of the device, and the device's authentication is successful. Otherwise, this proof is rejected, and the device's authentication is unsuccessful.

5. Security analysis

5.1. Zero-knowledge authentication

Firstly, we analyze three important properties of LZIA. Namely, completeness, soundness and zero-knowledge.

5.1.1. Completeness

Theorem 1. *In LZIA, for an honest embedded device, if it owns the correct private key, the AS can successfully authenticate the identity of the device.*

Proof. On the premise that the embedded device and the AS are honest and do not violate the interaction protocol, the device can obviously calculate the correct c_q value in each round of inquiry and return the corresponding information to the AS to prove its identity. The reason is that the device knows the private key sk , secret value $\{z_1, z_2\}$, and the q value selected by the AS in the interaction. In other words, for any challenge value q of the AS, the response made by the embedded device can always meet the following conditions.

(1) When $q = 1$, the following equation holds.

$$\begin{aligned} T_{c_1}(q) \bmod p &= T_{sk \times z_1 \times rand_3 \times H(ID \| TS)}(q) \bmod p \\ &= T_{H(ID \| TS)}(T_{sk \times z_1}(T_{rand_3}(q))) \bmod p \\ &= T_{H(ID \| TS)}(Y_1(Q)) \bmod p \\ &= vk_1 \end{aligned} \quad (9)$$

(2) When $q = 2$, the following equation holds.

$$\begin{aligned} T_{c_2}(q) \bmod p &= T_{sk \times z_2 \times rand_3 \times H(ID \| TS)}(q) \bmod p \\ &= T_{H(ID \| TS)}(T_{sk \times z_2}(T_{rand_3}(q))) \bmod p \\ &= T_{H(ID \| TS)}(Y_2(Q)) \bmod p \\ &= vk_2 \end{aligned} \quad (10)$$

(3) When $q = 3$, the following equation holds.

$$\begin{aligned} T_{c_3-2 \times H(ID \| TS)}(q) \bmod p + T_{c_3}(q) \bmod p &= T_{sk \times rand_3 - H(ID \| TS)}(q) \bmod p + T_{sk \times rand_3 + H(ID \| TS)}(q) \bmod p \\ &= 2 \times T_{sk \times rand_3}(q) \times T_{H(ID \| TS)}(q) \bmod p \\ &= 2 \times T_{sk}(Q) \times T_{H(ID \| TS)}(q) \bmod p \\ &= 2 \times pk(Q) \times T_{H(ID \| TS)}(q) \bmod p \\ &= vk_3 \end{aligned} \quad (11)$$

The embedded device generates the above c_1 , c_2 and c_3 based on the private key sk , secret value $\{z_1, z_2\}$, random number $rand_3$, and other data. The AS generates vk_1 , vk_2 , and vk_3 according to the chaotic polynomials $\{Y_1(x), Y_2(x)\}$, identifier ID , public key $pk(x)$, and other data. Assume that the embedded device and the AS generate the proof key $\{z_1, z_2\}$, proof c_q , and verification key vk_q according to the normal execution process. In that case, the AS can successfully authenticate the device identity for any number of queries. To sum up, if the device provides correct data, the probability of successful verification by the AS is $Pr = 1$. Therefore, the protocol meets the completeness.

5.1.2. Soundness

Theorem 2. *In LZIA, if the embedded device does not have the correct private key, it cannot pass the verification of the AS.*

Proof. According to the inverse negative proposition of Theorem 2, if the embedded device can pass the verification of the AS, the device must have the correct private key. We assume that the AS can accept the malicious device without knowing the private key. It shows that

the malicious device can respond correctly to all the challenge values q in the interaction process. Specifically, the response of the malicious device is c_1' , c_2' and c_3' , which meet the following conditions:

$$\begin{cases} c_1' = c_1 = sk \times z_1 \times rand_3 \times H(ID \| TS) \\ c_2' = c_2 = sk \times z_2 \times rand_3 \times H(ID \| TS) \\ c_3' = c_3 = sk \times rand_3 + H(ID \| TS) \end{cases} \quad (12)$$

The ways to satisfy the above conditions are twofold. One is that all responses of malicious devices randomly collide successfully, but the probability is negligible. Second, the malicious device knows the fake device's private key sk and the proof key $\{z_1, z_2\}$. However, in the whole authentication phase, the private key sk and the proof key $\{z_1, z_2\}$ are never transmitted and shared. Therefore, malicious devices cannot steal private information on the channel. Considering that Chebyshev polynomials have one-way uniqueness, if and only if the malicious device must provide the correct private key sk to pass verification. Therefore, malicious devices cannot use other values to pass the verification. They can only derive the private key sk from shared data involving privacy information such as chaotic polynomials $\{Y_1(x), Y_2(x)\}$ and proofs c_q under different challenge values q . Among them, on the premise that x , $Y_1(x)$, and $Y_2(x)$ are known, the malicious device needs to derive the value $sk \times z_1$ from Y_1 and the value $sk \times z_2$ from Y_2 . In other words, the malicious device can solve the chaotic map-based discrete logarithm problem, but this is an NP-hard problem. Since proof c_q combines a random number $rand_3$, each round of proof c_q is generated with a different random number $rand_3$. Therefore, when the finite field Z_p is large enough, it is difficult for malicious devices to deduce the private key sk from the proof c_q directly. The malicious device can only calculate the private key sk by solving the chaotic map-based discrete logarithm problem, deducing the random number $rand_3$ from the shared chaotic value Q , and combining multiple proofs c_q .

To sum up, the malicious device can only generate effective proof if it can successfully collide randomly many times or solve the chaotic map-based discrete logarithm problem in polynomial time. However, the probability of successful multiple random collisions is negligible. The chaotic map-based discrete logarithm problem is an NP-hard problem that cannot be solved in polynomial time. Therefore, if the device does not have the correct private key, it cannot pass the identity verification of the AS. The protocol meets the soundness.

5.1.3. Zero-knowledge

Theorem 3. *LZIA meets the requirement of zero knowledge. Namely, either the AS or other malicious devices cannot know any valuable information about the honest embedded devices.*

Proof. We prove the zero-knowledge of LZIA by constructing a simulator sim . In polynomial time, it cannot be distinguished from the actual authentication process. The simulator sim can allow the verifier to communicate any number of times. Therefore, the simulator sim can restart the entire authentication protocol until the authentication data of the embedded device is guessed correctly. The simulator sim is used to generate legal interactive content. In the following simulation process, the simulator sim can communicate with the AS normally:

Step 1: The simulator sim randomly selects two integers $\{x_1', x_2'\}$ from the finite field Z_p and selects a registered ID as the hash value H . Then, the simulator sim calculates the commitment value $C' = H(x_1' \| x_2')$ and sends it to the AS.

Step 2: The verifier hashes the commitment value C' , identifier TID and a random number $rand_1$. Then, the simulator sim replaces the generated hash value H with the hash value to be calculated by the verifier.

Step 3: The simulator sim randomly selects a challenge value $q' \in \{1, 2, 3\}$ and random number $rand_3 \in Z_p$, and then generates a chaotic value $Q' = T_{rand_3}(q')$.

Step 4: The simulator *sim* selects three random numbers $z_1', z_2', sk' \in Z_p$ and calculates them according to Eq. (13).

$$\begin{cases} c_1' = z_1' \times rand_3 \times H \\ c_2' = z_2' \times rand_3 \times H \\ c_3' = sk' \times rand_3 + H \end{cases} \quad (13)$$

Step 5: The verifier returns a challenge value q .

Step 6: If $q' \neq q$, the simulator *sim* does not process and returns to Step 3 to re-execute the process. If $q' = q$, but the proof c_q' provided cannot pass the verification, the simulator *sim* returns to Step 3 and re-executes the process. If $q' = q$ and the proof c_q' provided passes the verification, then the simulator *sim* records the current round of interaction.

Through the above steps, if the simulator *sim* can correctly respond to all the challenge values of the verifier, it is proved that the AS cannot distinguish the proof provided in polynomial time. The reason is that both the data returned by the simulator *sim* and the actual data can be verified successfully. The AS cannot obtain more privacy information from the two known data in the polynomial time. This makes the AS unable to distinguish whether the device or simulator *sim* provides the data. Therefore, the authentication phase of the simulator *sim* and the actual prover is indistinguishable in polynomial time. The AS cannot obtain any valuable information. Therefore, the protocol satisfies zero-knowledge.

5.2. Anti-attack capability analysis

5.2.1. Informal security analysis

LZIA can effectively defend against these common attacks mentioned in Section 3, and meet the security requirements of anti-attack capability. The specific analysis is as follows:

- (1) **Replay Attack:** In the LZIA, the authentication interaction between the embedded device and the AS contains a random number $rand_3$. In each verification process, the data changes with the random number, making it impossible for adversaries to know the random number in polynomial time due to the chaotic map-based discrete logarithm problem. Any re-issued historical data cannot pass identity verification. Additionally, the AS can detect the adversary's malicious behavior by checking the timestamp of the data interaction, making a replay attack almost impossible.
- (2) **Counterfeit Attack:** LZIA uses Chebyshev polynomials to calculate the public key $pk(x) = T_{sk}(x) \bmod p$. The complexity of the Chebyshev discrete logarithm problem is high enough that an adversary cannot calculate the device's private key by intercepting the public key information. In addition, there is a random number $rand_3$ in each verification process. The adversary cannot calculate the device's private key by eavesdropping on data, so he cannot forge the identity of embedded devices. A zero-knowledge proof-based authentication protocol LZIA ensures that an adversary cannot successfully verify it.
- (3) **Sensor Node Capture Attack:** We assume that the adversary obtains the private information of some legitimate devices in the network. However, the public and private keys of each device are different, and the private key is obtained through the random number $\{x_1, x_2\}$ and ordered sequence $\{s_1, s_2\}$ calculation. The random number $\{s_1, s_2\}$ is not transmitted on the channel, and the ordered sequence $\{s_1, s_2\}$ is deleted from the memory after the private key is generated. Furthermore, the AS periodically updates the public and private keys before their validity period expires. Therefore, even if an adversary manages to capture a small amount of a device's private information, they will not be able to use it to launch a successful attack on the entire network.

- (4) **Man-in-the-middle Attack:** The authentication scheme LZIA in this paper adopts the zero-knowledge proof protocol, which satisfies the zero-knowledge property. Even if an adversary intercepts the communication information between the device and the AS, they will not be able to obtain the device's private information from the intercepted data. Furthermore, any tampered data will not be able to pass the authentication.
- (5) **Server Insider Attack:** The authentication phase of this scheme needs to verify the zero-knowledge proof of the embedded device. In the authentication phase, the secret data $\{x_1, x_2, z_1, z_2, sk\}$ used to calculate the proof has no interaction. Therefore, even if the adversary is an insider of the AS, he cannot get the secret data and impersonate any device.
- (6) **Privileged Insider Attack:** We assume that the adversary is an insider developer of the device and can access the storage information $\{ID, pk(x), Y_1(x), Y_2(x)\}$ of each device. Since the login password x_1 and secret random number x_2 are not stored in the device memory, the adversary cannot obtain the x_1 and x_2 of the device and pass authentication. Therefore, the adversary cannot proceed with further identity verification. The scheme is resilient against privileged internal attacks.
- (7) **Guessing Attack:** To achieve illegal authentication and access, adversaries must obtain the secret data $\{x_1, x_2\}$ and $\{z_1, z_2, sk\}$ of a legitimate device. However, the characteristics of the secure hash function and Chebyshev polynomial make the adversaries can only exhaustively crack the secret data. The adversaries must guess five secret data with finite field Z_p simultaneously to pass the authentication, which has a small probability. Moreover, LZIA updates the secret key periodically, which makes it difficult for adversaries to carry out guessing attacks.

5.2.2. Formal security analysis

In this section, we conducted a formal analysis of LZIA using the widely-used random-or-real (ROR) model firstly. Assume that there is a polynomial-time adversary $\mathcal{A}$ who can access all messages between device D and the AS, as well as all public parameters. In LZIA, adversary $\mathcal{A}$ can make the following oracle queries in polynomial time to compromise the security of the protocol:

- (1) *Send*(AS, m). The query simulates that adversary $\mathcal{A}$ masquerades as an honest device and sends message m to AS.
- (2) *Send*(D , m). The query simulates that adversary $\mathcal{A}$ masquerades as an AS and sends message m to the device.
- (3) *Execute*(D , m). The query simulates that adversary $\mathcal{A}$ eavesdrops all messages between the device and AS on the communication channel.
- (4) *Corrupt*(D , AS). The query simulates that adversary $\mathcal{A}$ accesses secret information in the memory of the device or AS.
- (5) *Test*(D). The query measures the semantic security of the device's proof. After the *Test*(D) query, D tosses a coin b . If $b = 1$, D returns the real proof to $\mathcal{A}$. If $b = 0$, it returns a random proof of the same length as the real proof.

We evaluate the security of LZIA through the game G_i ($i = 0, 1, 2, 3, 4, 5$). In the game, $\mathcal{A}$ can initiate multiple *Test*(D) queries against device D . After receiving the queries, device D tosses a coin b (result is 0 or 1). If adversary $\mathcal{A}$ correctly guesses the value of b , it wins the game. The advantage probability $Adv_D(\mathcal{A})$ of adversary $\mathcal{A}$ in breaking the security of our scheme is defined as

$$Adv_D(\mathcal{A}) = |2 \Pr[Succ_{\mathcal{A}}] - 1| < \epsilon \quad (14)$$

where $\Pr[Succ_{\mathcal{A}}]$ represents the probability of adversary $\mathcal{A}$ winning the game, ϵ is a very small negligible value.

Theorem 4. The probability of adversary $\mathcal{A}$ winning the game G_i in probabilistic polynomial time is negligible. To break the semantic security of LZIA, adversary $\mathcal{A}$ makes hash oracle queries, *Send* queries and *Execute* queries at most q_h , q_s and q_e times, respectively. Where q_h represents the

number of hash oracle queries, q_s represents the number of Send queries, q_e represents the number of Execute queries. Then, we have

$$Adv_D(\mathcal{A}) = \frac{2q_h^2}{|hash|} + \frac{(q_s + q_e)^2}{p} + 2\left(\frac{q_s}{|hash|} + Adv_{CMDLP}(\mathcal{A})\right) \quad (15)$$

where $|hash|$ represents the size of hash query, $Adv_{CMDLP}(\mathcal{A})$ represents the probability of $\mathcal{A}$ solving CMDLP in polynomial time.

Proof. We deduce the probability of adversary $\mathcal{A}$ breaking the scheme through the game G_i ($i = 0, 1, 2, 3, 4, 5$). Let $\Pr[Succ_i]$ represent the probability of adversary $\mathcal{A}$ guessing the value of coin b in the game G_i .

(a) G_0 : The game simulates the real attack scenario, and we obtain

$$Adv_D(\mathcal{A}) = |2\Pr[Succ_0] - 1| \quad (16)$$

(b) G_1 : The game simulates adversary $\mathcal{A}$'s replay attack and man-in-the-middle attack by executing $Send(D, m)$, $Send(AS, m)$ and $Execute(D, m)$ queries. Adversary $\mathcal{A}$ obtains the messages $\{C', q, c_1, c_2, c_3, Q, TS\}$ through eavesdropping, but it cannot calculate the private key sk , secret parameter $\{x_1, x_2, z_1, z_2\}$ and random number $rand_3$ from these messages. Additionally, outdated proofs cannot pass verification. Therefore, at the end of the game, Adversary $\mathcal{A}$ cannot judge whether the parameter returned by device D through the $Test(D)$ query is a real proof or a random proof of equal length. Therefore, compared with the real attack scenario, adversary $\mathcal{A}$ does not have an increased advantage in this game, so we have

$$\Pr[Succ_1] = \Pr[Succ_0] \quad (17)$$

(c) G_2 : The game removes two collision cases in G_1 , and simulates adversary $\mathcal{A}$'s counterfeit attack.

C_1 : According to the birthday paradox [33], the collision probability in the $hash$ query output is less than or equal to $\frac{q_h^2}{2|hash|}$.

C_2 : The selected random number collides, and its probability is less than or equal to $\frac{(q_s + q_e)^2}{2p}$.

Unless the above cases, G_2 and G_1 are indistinguishable. Therefore, we have

$$\Pr[Succ_2] - \Pr[Succ_1] \leq \frac{q_h^2}{2|hash|} + \frac{(q_s + q_e)^2}{2p} \quad (18)$$

(d) G_3 : The game eliminates adversary $\mathcal{A}$'s direct guess of proof c_1 , c_2 or c_3 in G_2 without the necessity of making a $hash$ query, thus it simulates adversary $\mathcal{A}$'s guessing attack. The probability of the situation is less than or equal to $\frac{q_s}{|hash|}$. Therefore, we have

$$\Pr[Succ_3] - \Pr[Succ_2] \leq \frac{q_s}{|hash|} \quad (19)$$

(e) G_4 : The game modifies G_3 to simulate the server insider attack. In the game, adversary $\mathcal{A}$ executes the $Corrupt(D, AS)$ query to obtain the parameters of the device stored in AS. With knowledge of $\{ID, Y_1(x), Y_2(x), pk(x)\}$, adversary $\mathcal{A}$ must know $\{sk, z_1, z_2\}$ to obtain the device's proof. But to obtain $\{sk, z_1, z_2\}$ from $pk(x) = T_{sk}(x)$, $Y_1(x) = T_{z_1 \times sk}(x)$ and $Y_2(x) = T_{z_2 \times sk}(x)$, adversary $\mathcal{A}$ needs to solve the CMDLP. Therefore, comparing the difference between the game and G_3 , we can obtain

$$\Pr[Succ_4] - \Pr[Succ_3] \leq Adv_{CMDLP}(\mathcal{A}) \quad (20)$$

(f) G_5 : The game modifies G_4 to simulate adversary $\mathcal{A}$'s sensor node capture attack and privileged insider attack. In the game, adversary $\mathcal{A}$ executes the $Execute(D, m)$ query to obtain the parameters on the communication channel and the $Corrupt(D, AS)$ query to obtain the secret parameters of device D . With knowledge of $\{sk, z_1, z_2\}$, $\mathcal{A}$ must compute $ID = H(H(C) \| TID \| rand_1)$ to obtain the device's proof $c_1 = sk \times z_1 \times rand_3 \times H(ID \| TS)$, $c_2 = sk \times z_2 \times rand_3 \times H(ID \| TS)$ and $c_3 = sk \times rand_3 \times H(ID \| TS)$. However, to extract x_1 and x_2 from $C = H(x_1 \| x_2)$, adversary $\mathcal{A}$ needs to crack the one-way property of the

hash function. Therefore, comparing the difference between the game and G_4 , we have

$$\Pr[Succ_5] - \Pr[Succ_4] \leq \frac{q_h^2}{2|hash|} \quad (21)$$

In the above games, adversary $\mathcal{A}$ gains no advantage to guess the correct bit b , we get

$$\Pr[Succ_5] = \frac{1}{2} \quad (22)$$

According to Eqs. (16) and (17), we get

$$\frac{1}{2} Adv_D(\mathcal{A}) = \left| \Pr[Succ_0] - \frac{1}{2} \right| = \left| \Pr[Succ_1] - \frac{1}{2} \right| \quad (23)$$

According to Eqs. (22) and (23), we get

$$\frac{1}{2} Adv_D(\mathcal{A}) = |\Pr[Succ_1] - \Pr[Succ_5]| \quad (24)$$

According to Eqs. (18)~(21) and the triangular inequality, we have

$$\begin{aligned} & |\Pr[Succ_1] - \Pr[Succ_5]| = |\Pr[Succ_1] - \Pr[Succ_2] \\ & + \Pr[Succ_2] - \Pr[Succ_3] \\ & + \Pr[Succ_3] - \Pr[Succ_4] \\ & + \Pr[Succ_4] - \Pr[Succ_5]| \\ & \leq |\Pr[Succ_1] - \Pr[Succ_2]| \\ & + |\Pr[Succ_2] - \Pr[Succ_3]| \\ & + |\Pr[Succ_3] - \Pr[Succ_4]| \\ & + |\Pr[Succ_4] - \Pr[Succ_5]| \\ & \leq \frac{q_h^2}{|hash|} + \frac{(q_s + q_e)^2}{2p} + \frac{q_s}{|hash|} + Adv_{CMDLP}(\mathcal{A}) \end{aligned} \quad (25)$$

Finally, according to Eqs. (24) and (25), we get Eq. (15).

Hence, Theorem 4 is proved. The above proof process means that the probability of adversary $\mathcal{A}$ winning in the game is negligible. Therefore, LZIA is secure under the ROR model.

5.2.3. Penetration testing

We build an authentication environment using Python programming, which consists of two honest devices, a malicious device, and the AS. Then, we choose similar lightweight authentication schemes for resource-constrained IoT devices, such as LAKA [29] based on elliptic curve cryptography, SRAM [30], and EEAS [32] based on Chebyshev polynomials. To compare and analyze the schemes' anti-attack capability, we use a malicious device to simulate seven common types of attacks against the system.

(1) Replay Attack: The malicious device listens and records messages between the honest device and the AS on the wireless channel. Then, it re-transmits the information to AS.

(2) Counterfeit Attack: The malicious device uses equations to generate new secret data. Then, it combines the communication data between the honest device and the AS, and authenticates with the AS.

(3) Sensor Node Capture Attack: The malicious device physically accesses the first honest device to obtain its secret data and listens to the communication between the second honest device and AS. Then, it uses the first device's data to verify the identity of the second device.

(4) Man-in-the-middle Attack: The malicious device intercepts and tampers with the communication data between the honest device and the AS through the wireless channel.

(5) Server Insider Attack: The malicious device physically accesses the device registration data stored in the database. Then, it uses the data to establish authentication communication with the AS.

(6) Privileged Insider Attack: The malicious device physically accesses an honest device to obtain private data, and uses the data to authenticate with AS.

(7) Guessing Attack: The malicious device uses brute force to guess various secret data and authenticate with AS until the identity is successfully verified.

According to the penetration testing, we obtain the results of anti-attack capability comparison in LZIA, EEAS, LAKA, and SRAM. As

Table 3
Anti-attack capability comparison.

Attack type	LZIA	EEAS	LAKA	SRAM
Replay Attack	✓	✓	✓	✓
Counterfeit Attack	✓	✓	✓	✓
Sensor Node Capture Attack	✓	✓	✓	✓
Man-in-the-middle Attack	✓	✓	✓	✓
Server Insider Attack	✓	✓	×	✓
Privileged Insider Attack	✓	✓	×	✓
Guessing Attack	✓	×	×	×

shown in Table 3, LZIA, EEAS, LAKA, and SRAM can defend against replay attacks, counterfeit attacks, sensor node capture attacks, and man-in-the-middle attacks. However, LZIA's secret data are generated locally on the device, and it uses a zero-knowledge proof protocol to ensure the security of the secret data. Therefore, it can defend against server insider attacks. Then LZIA's secret data $\{x_1, x_2\}$ is not stored in device memory, and the malicious device is unable to obtain $\{x_1, x_2\}$ and successfully passes the authentication. Therefore, it can defend against privileged insider attacks. LZIA needs to guess more secret data correctly at the same time, and updates regularly secret data. Therefore, LZIA can defend against guessing attacks.

Compared with LZIA, in terms of server insider attacks, the LAKA calculates and distributes the device's secret key by the AS during the registration phase. If the malicious device is an insider of the AS, it can access the secret data of any device and is unable to defend against server insider attacks. In terms of privileged insider attacks, LAKA stores the secret data calculated by the AS locally in the device memory and does not consider the protection and security of the data during storage. If the malicious device is an internal developer of the embedded device, it has the privilege to access the identity data in the memory and knows the secret data of any device. Therefore, LAKA cannot defend itself against privileged insider attacks. In terms of guessing attacks, LAKA, EEAS, and SRAM have a small amount of secret data, and the secret data cannot be updated regularly. When the malicious device correctly guesses the secret data through brute force, it can achieve illegal authentication and access. Therefore, LAKA, EEAS, and SRAM still have security risks in the face of guessing attacks. Therefore, LZIA is a more effective and comprehensive scheme in defending against the seven typical attacks compared with other authentication schemes.

6. Experimental analysis

6.1. Simulation analysis

6.1.1. Simulation parameters and performance parameters

To evaluate the performance of LZIA, we conducted tests on a desktop with a 64-bit Win11 operating system and an AMD Ryzen 7 5800U as its CPU. We implemented an authentication system using Python, consisting of embedded devices, an AS and an adversary. The initialization of experiment parameters is meticulously conducted based on existed literature and our experimental simulations. Specifically, we choose the expected security size of finite field as 1024 bits according to the [18]. To minimize the computation cost, we choose the actual size of the finite field per round of the verification process as 160 bits through our experimental simulations. Additionally, due to SHA1's robust security properties, efficient computation, and wide acceptance in the cryptographic community, we choose SHA1 as the one-way hash function. Then, we study the computation, storage, and communication costs of LZIA, EEAS, LAKA, and SRAM. Namely, the bit length of the Chebyshev polynomial operation in EEAS and SRAM is 160 bits. In the LAKA, the elliptic curve is secp160r1, and its symmetric encryption is AES. Note that computation cost is defined as the total time spent in the authentication process for an embedded device or AS. The storage cost is defined as the memory size occupied by the data to be saved

Table 4
Computation cost of cryptographic elements.

Cryptographic element	Execution time on AMD Ryzen 7 5800U
Chebyshev polynomial	0.4259825 ms
SHA1	0.0016498 ms
AES encryption	0.0340414 ms
AES decryption	0.0100159 ms
Elliptic curve point multiplication	8.4059977 ms
Modular exponentiation	1.6130896 ms

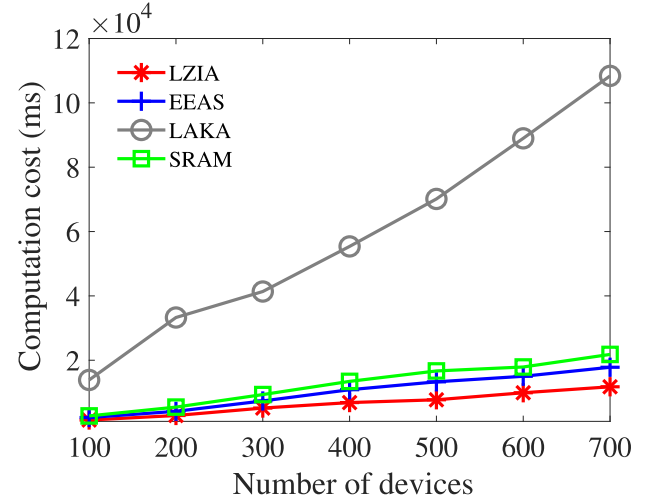


Fig. 6. Computation cost with different number of devices.

to achieve authentication successfully. The communication cost is the total memory size of all data transmitted between the device and the AS in the authentication phase.

6.1.2. Simulation performance analysis

In this section, we evaluate the computation, communication, and storage costs of LZIA, EEAS, LAKA, and SRAM.

First, we construct different cryptographic elements, including Chebyshev polynomial, SHA1, AES encryption/decryption, elliptic curve point multiplication, and modular exponentiation. Each cryptographic element is performed 100 times, and the average results are shown in Table 4. As shown in Table 4, we can conclude that Chebyshev polynomial operations are more efficient than the elliptic curve point multiplication operation and modular exponentiation operation.

The number of devices is selected as 100, 200, 300, 400, 500, 600, and 700, respectively. Then, we analyze the scheme computation cost with the different number of devices. As shown in Fig. 6, as the number of devices increases, the embedded devices' average computation cost of LZIA, EEAS, LAKA, and SRAM gradually increases. However, the increase in computation cost of LZIA is gradual and significantly lower compared to other schemes. This is because as the number of devices increases, both the device and the AS need to perform more encryption operations, but LZIA requires fewer computations than other schemes. This led to longer authentication execution time for multiple embedded devices. Therefore, the computation cost of all authentication schemes increases gradually. LAKA involves heavyweight encryption operations during a round of authentication for each device. It requires performing five elliptic curve point multiplication operations, two AES encryption/decryption operations, six modular exponentiation operations, and nine hash operations. This authentication scheme has the longest execution time, leading to the highest computation cost among the compared schemes. EEAS and SRAM realize authentication based on lightweight encryption elements such as Chebyshev polynomial

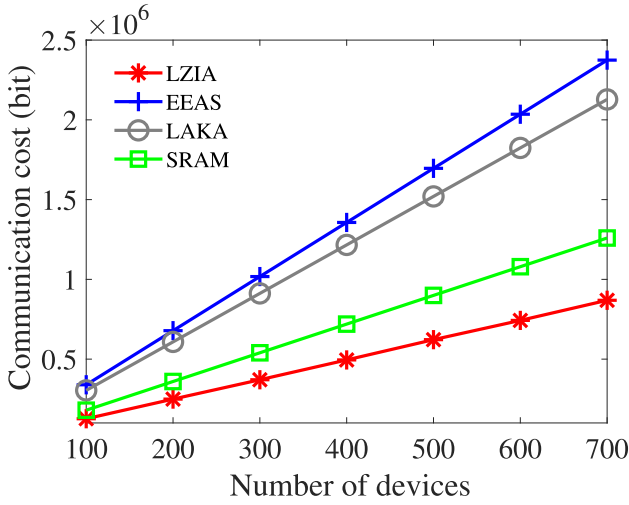


Fig. 7. Communication cost with different number of devices.

and hash operations. Their execution time is significantly less than the time taken for complex operations such as elliptic curve point multiplication, modular exponentiation, and other similar operations. Among them, EEAS needs to perform a total of seven Chebyshev polynomial operations and twelve hash operations. SRAM needs to perform a total of twenty-one Chebyshev polynomial operations and six hash operations. Therefore, their computation cost is significantly lower than that of LAKA. However, LZIA only requires performing one Chebyshev polynomial operation and two hash operations on the device and needs to execute three Chebyshev polynomial operations and three hash operations on the AS. It needs to perform a total of four Chebyshev polynomial operations and five hash operations. The execution time of the authentication scheme is the minimum. Therefore, LZIA is superior to other authentication schemes in computation cost.

Then, we compare the communication cost of each scheme and analyze the communication cost with the different number of devices. As shown in Fig. 7, as the number of devices increases, the embedded devices' communication cost of LZIA, EEAS, LAKA, and SRAM gradually increases, while LZIA is lower than other schemes. As the number of devices increases, the number of interactions that the AS needs to perform and the amount of authentication data transferred also increase, leading to an increase in the communication cost of all authentication schemes. In the process of a round of authentication for each device, EEAS requires introducing a trusted party to achieve registration, resulting in a large amount of data transmitted in the registration phase. It needs to send two 160 bits Chebyshev polynomial values, seven 160 bits hash values, two 160 bits XOR values, and two 160 bits random numbers. In the authentication phase, it needs to transmit two 160 bits Chebyshev polynomial values and five 160 bits hash values. LAKA uses the AS to generate the secret key, unique identification, and other data of the device and sends the generated data to the device, resulting in a large number of data transmitted in the registration phase. It needs to send a 160 bits elliptic curve, elliptic curve base point, large prime number, hash value, private key, and two 160 bits random numbers. In the authentication phase, it needs to transmit a total of twelve 160 bits values. Therefore, EEAS and LAKA have a terrible effect on communication costs. In the SRAM, each device needs to transmit a 13 bits timestamp and four 160 bits values in the registration phase and two 13 bits timestamps and seven 160 bits values in the authentication phase. However, LZIA only needs to transmit three 160 bits values in each device's registration phase and send a 2 bits challenge value, a 13 bits timestamp and three 160 bits values in the authentication phase. Therefore, LZIA is superior to other authentication schemes on communication costs.

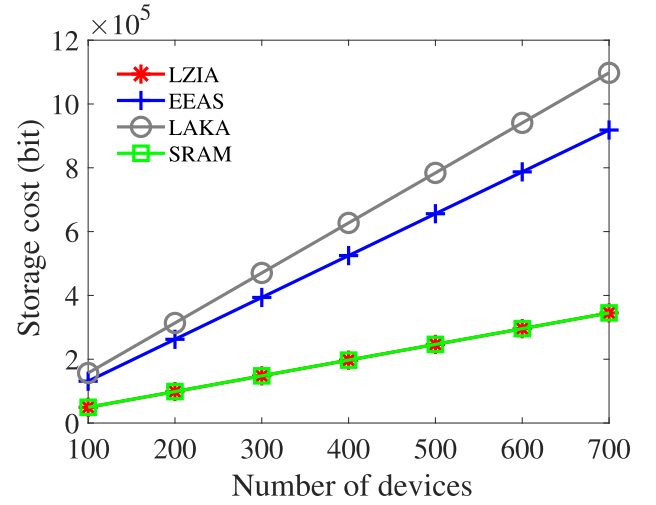


Fig. 8. Communication cost with different number of devices.

Finally, we compare the storage cost of each scheme and analyze the cost with the different number of devices. As shown in Fig. 8, as the number of devices increases, the embedded devices' storage cost of LZIA, EEAS, LAKA, and SRAM gradually increases, while LZIA and SRAM are consistent and significantly lower than EEAS and LAKA. As the number of devices increases, the AS needs to store more device data, resulting in an increase in the storage cost of all authentication schemes. In EEAS, each device needs to save seven 160 bits values in the memory in the registration phase. In LAKA, each device needs to save five 160 bits values and three 256 bits values in the device memory in the registration phase. In LZIA, each device only needs to store the private key sk , timestamp ΔT , and secret data z_1 and z_2 in the device memory, so it only needs to store a total of 493 bits values. Similarly, SRAM needs to store three 160 bits values and a 13 bits timestamp, and its storage cost is the same as that of LZIA. Therefore, LZIA and SRAM are superior to EEAS and LAKA in storage costs.

In summary, we compare LZIA, EEAS, LAKA, and SRAM. LZIA can effectively reduce the computation, communication, and storage costs of the embedded device.

6.2. Hardware implementation and analysis

This section describes the setup of a hardware test platform and the integration of the authentication protocol into it. Based on the hardware test platform, we compare the computation costs of LZIA, EEAS, LAKA, and SRAM.

6.2.1. Hardware test platform construction

In the experiment, the parameters are the same as presented in Section 6.1.1, and the hardware configuration and interaction diagram are shown in Fig. 9. An STM32 development board simulates the embedded device in the experiment. The STM32 development board has a STM32F103ZET6 chip which is a 32-bit ARM Cortex-M3 CPU with a maximum clock frequency of 72 MHz and a data bus width of 32 bits. It also has a 512 KB capacity flash memory and 48 KB SRAM memory. The embedded device connects to the AS through the HC-05 Bluetooth module to realize the data communication between the device and the AS. During the authentication process, the STM32 MCU generates authentication information and stores data such as the private key in the SRAM. Then it sends the information to the AS via Bluetooth wireless signals. Upon receiving the signals, the AS processes them to complete the authentication. To observe the intricate details of the embedded device's protocol execution process on the computer, we employ the XCOM software as the application server. This software can

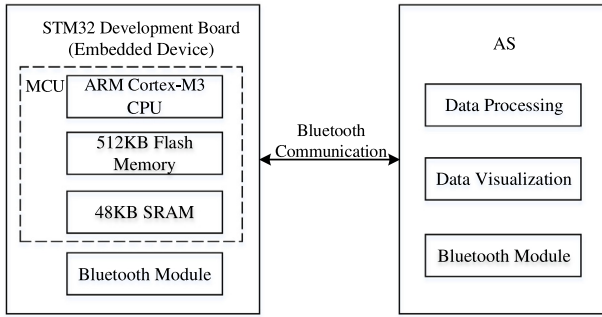


Fig. 9. Embedded device hardware configuration and interaction diagram.

Table 5
Computation cost of cryptographic elements.

Cryptographic element	Execution time on STM32 development board
Chebyshev polynomial	556.690 ms
SHA1	2.939 ms
AES encryption	11.136 ms
AES decryption	10.467 ms
Elliptic curve point multiplication	13847.501 ms
Modular exponentiation	1398.390 ms

enable us to monitor each step of the execution process of the device and obtain insights into the protocol execution process.

After constructing the hardware testing platform, we use the C language to implement the LZIA, EEAS, LAKA, and SRAM in the integrated development environment of Keil MDK5. Then, we study the intricacies of the authentication phase of both the embedded device and the AS and measure various performance metrics. Finally, we compare the computation cost of each scheme based on the performance metrics obtained.

6.2.2. Performance comparison

First, we construct different cryptographic elements through Keil MDK5 software. Each cryptographic element is performed 100 times, and the average results are shown in Table 5. From Table 5, we summarize that Chebyshev polynomial operations are more efficient than the elliptic curve point multiplication operation and modular exponentiation operation. In addition, we note that the size relationship of the execution time of each cryptographic element in Table 5 is consistent with that in Table 4. But the execution time of each cryptographic element in Table 5 is greater than that of the corresponding cryptographic element in Table 4. The reason is that the STM32 development board has far less computing power than computers, and the programming of cryptographic elements in C language is more complex than in Python.

The communication and storage costs of the authentication scheme are independent of its execution environment. Therefore, the communication and storage costs on the hardware testing platform are consistent with the simulation results of the scheme. Moreover, each device only needs to perform the registration phase once. Therefore, we only compare the computation cost of each authentication phase in the hardware platform environment. Let the symbols T_c , T_h , T_s , T_m , T_e , and ΔT denote the time for executing a Chebyshev polynomial operation, a one-way hash function operation, an AES encryption operation, an AES decryption operation, an elliptic curve point multiplication operation, a modular exponentiation operation, and other operations, respectively. As shown in Table 6, when the embedded device is certified, the LZIA's computation cost and total cost in the device are less than those of EEAS, LAKA, and SRAM. The reason is that LZIA avoids the complex modular exponentiation operation and adopts the efficient Chebyshev polynomial operation in the authentication phase. It needs

to perform two hash operations, one Chebyshev polynomial operation, and other operations on the embedded device. Then, it needs to perform three hash operations, three Chebyshev polynomial operations, and other operations on the AS. Although EEAS and SRAM also use Chebyshev polynomials with low computation costs, the total number of hashes and Chebyshev polynomials that need to be performed on the device side are higher than LZIA, their computation costs on the embedded device and total computation costs are higher than LZIA. Furthermore, LAKA entails several intricate operations, such as elliptic curve point multiplication and modular exponentiation, resulting in its computation cost being the highest.

In summary, the communication and storage costs of LZIA in computer simulation and embedded devices are consistent. Because the communication and storage costs mainly depend on the specific authentication protocol, rather than the executing platform. And in LZIA, the number and size of message transmissions, as well as the amount of stored data, are fixed. Then the computation cost of the computer simulation is less than that of the embedded devices, but their computation cost trends are consistent since the computational capability of the embedded devices is weaker than that of the computer. Therefore, the computation cost of embedded devices is much higher than that of computer simulation when the same authentication protocol is running. However, the operation steps and order of the executed protocol and the proportion of time consumed in each part of the protocol are the same regardless of the different experiment platforms. Therefore, their trends of computation cost are consistent.

7. Conclusion

This paper introduces a lightweight zero-knowledge authentication scheme (LZIA) designed specifically for IoT-embedded devices to avoid the high costs of computation, communication, and storage required in traditional complex security authentication schemes. Firstly, we propose a lightweight registration method. Specifically, we bind identity information to a unique identifier and achieve device identity allocation through hash commitment. Secondly, we introduce a method for secret key management where we utilize the device identity information and Chebyshev polynomial chaotic map to simplify the device's secret key structure. We also propose a secret key generation equation, an updated equation, and a secret key management method. Finally, we employ a lightweight zero-knowledge authentication protocol. By integrating all intermediate data into a Chebyshev polynomial for unified verification, the proof generation equation and the proof verification equation based on the three challenge values are improved. As such, a lightweight interactive zero-knowledge authentication process is presented. Experimental results demonstrate that LZIA can effectively reduce computation, communication, and storage costs while ensuring security in both simulated and actual hardware environments. It outperforms EEAS, LAKA, and SRAM in both security and performance.

The LZIA has been successfully deployed on the embedded device with a clock frequency of 72MHz, capacity flash memory of 512 KB, and SRAM of 48 KB, whose limited computational resources are less than most IoT devices on the current market. The LZIA can be applied to most IoT devices with a few specific adaptations. Therefore, the LZIA is lightweight, flexible, has excellent scalability, and is adaptable to various IoT devices. In the future, we will use some professional tools such as AVISPA to do the formal analysis of LZIA and find potential problems or attacks. Additionally, we will also optimize the LZIA protocol for wearable IoT devices for smart healthcare, and develop the corresponding embedded software. And we will also study the effect of network delay and device mobility on the performance of the authentication protocol, and carry out application experiments of the scheme to further improve the efficiency and applicability of LZIA.

Table 6

Comparison of computation cost in authentication stage.

Scheme	Embedded device	AS	Total cost
LZIA	$2T_h + T_c + \Delta T \approx 902.648$ ms	$3T_h + 3T_c + \Delta T \approx 2170.913$ ms	$5T_h + 4T_c + \Delta T \approx 3073.561$ ms
EEAS	$7T_h + 5T_c + \Delta T \approx 2862.831$ ms	$5T_h + 2T_c + \Delta T \approx 1782.794$ ms	$12T_h + 7T_c + \Delta T \approx 4645.625$ ms
LAKA	$4T_h + 3T_m + T_s + 2T_c + \Delta T \approx 41546.166$ ms	$5T_h + 3T_m + T_s + 3T_c + \Delta T \approx 41556.297$ ms	$9T_h + 6T_m + 2T_s + 5T_c + \Delta T \approx 83102.463$ ms
SRAM	$4T_h + 10T_c + \Delta T \approx 6148.700$ ms	$2T_h + 11T_c + \Delta T \approx 7494.869$ ms	$6T_h + 21T_c + \Delta T \approx 13643.569$ ms

CRedit authorship contribution statement

Zhangquan Wang: Conceptualization, Writing – original draft. **Ji-axuan Huang:** Methodology. **Kelei Miao:** Experiment, Formal analysis. **Xiaowen Lv:** Writing – review & editing. **Yourong Chen:** Experiment, Formal analysis, Writing – original draft, Writing – review & editing, Supervision. **Bing Su:** Experiment, Formal analysis. **Liyuan Liu:** Writing – review & editing. **Meng Han:** Supervision, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

The research was funded by the “Ling Yan” Research and Development Project of Science and Technology Department of the Zhejiang Province of China under Grant No. 2023C03189, the Major Science and Technology Innovation Project of Hangzhou in the Zhejiang Province of China under Grant No. 2022AIZD0104, the Public Welfare Technology Application and Research Projects of Zhejiang Province of China under Grant No. LGF21F010004, and the National Key Research and Development Program of China under Grant No. 2023YFB2704400. All authors have read and agreed to the published version of the manuscript.

References

- [1] Yang Zhang, Xiaowen Lv, Yourong Chen, Tiaojuan Ren, Changchun Yang, Meng Han, FAWPA: A FAW attack protection algorithm based on the behavior of blockchain miners, *Sensors* 22 (13) (2022) 5032.
- [2] Mansoor Nasir, Khan Muhammad, Amin Ullah, Jamil Ahmad, Sung Wook Baik, Muhammad Sajjad, Enabling automation and edge intelligence over resource constraint IoT devices for smart home, *Neurocomputing* 491 (2022) 494–506.
- [3] Hao Chen, Yourong Chen, Zhenyu Xiong, Meng Han, Zaobo He, Banteng Liu, Zhangquan Wang, Zhenghua Ma, Prevention method of block withholding attack based on miners' mining behavior in blockchain, *Appl. Intell.* (2022) 1–19.
- [4] S.Q. Li, H.Y. Li, Application of intelligent filming equipment in medical health monitoring, *Eur. Rev. Med. Pharmacol. Sci.* 26 (16) (2022) 5660–5666.
- [5] Yourong Chen, Hao Chen, Yang Zhang, Meng Han, Madhuri Siddula, Zhipeng Cai, A survey on blockchain systems: Attacks, defenses, and privacy preservation, *High-Confid. Comput.* 2 (2) (2022) 100048.
- [6] Yang Zhang, Yourong Chen, Kelei Miao, Tiaojuan Ren, Changchun Yang, Meng Han, A novel data-driven evaluation framework for fork after withholding attack in blockchain systems, *Sensors* 22 (23) (2022) 9125.
- [7] Usman Khalil, Mueen-Uddin, Owais Ahmed Malik, Saddam Hussain, A blockchain footprint for authentication of IoT-enabled smart devices in smart cities: State-of-the-art advancements, challenges and future research directions, *IEEE Access* 10 (2022) 76805–76823.
- [8] Evangelina Lara, Leocundo Aguilar, Mauricio A. Sanchez, Jesus A. Garcia, *Sensors* 20 (2) (2022) 501.
- [9] Pawel Szalachowski, Password-authenticated decentralized identities, *IEEE Trans. Inf. Forensics Secur.* 16 (2021) 4801–4810.
- [10] Hui Zhang, Weixin Bian, Biao Jie, Shuwan Sun, BioF-TAP: An efficient method of template protection and two-factor authentication protocol combining biometric and PUF, *J. Intell. Fuzzy Systems* (2022) 1–17.
- [11] Kai Fan, Wei Jiang, Qi Luo, Hui Li, Yintang Yang, Cloud-based RFID mutual authentication scheme for efficient privacy preserving in IoV, 358, (1) 2021, pp. 193–209.
- [12] Wanxue Lin, Xuefei Zhang, Qimei Cui, Zhiwei Zhang, Blockchain based unified authentication with zero-knowledge proof in heterogeneous MEC, in: 2021 IEEE International Conference on Communications Workshops, (ICC Workshops), IEEE, 2021, pp. 1–6.
- [13] Shashank Agrawal, Peihan Miao, Payman Mohassel, Pratyay Mukherjee, PASTA: password-based threshold authentication, in: Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security, ACM, 2018, pp. 2042–2059.
- [14] Yuan Zhang, Chunxiang Xu, Hongwei Li, Kan Yang, Nan Cheng, Xuemin Shen, PROTECT: efficient password-based threshold single-sign-on authentication for mobile users against perpetual leakage, *IEEE Trans. Mob. Comput.* 20 (6) (2020) 2297–2312.
- [15] Xiang Zhang, Lina Yao, Chaoran Huang, Tao Gu, Zheng Yang, Yunhao Liu, DeepKey: A multimodal biometric authentication system via deep decoding gait and brainwaves, *ACM Trans. Intell. Syst. Technol.* 11 (4) (2020) 1–24.
- [16] Zhigang Gao, Zhichao Cheng, Wenjie Diao, Jianhui Zhang, Huijuan Lu, Identity authentication based on trajectory characteristics of mobile devices, *J. Syst. Archit.* 112 (2021) 101857.
- [17] Houzhen Wang, Xingwei Cai, Yan Guo, 5-pass zero-knowledge authentication scheme based on matrix completion problem, *J. Commun.* 42 (2021) 1–10.
- [18] Jian Song, Po-Wei Harn, Kazuya Sakai, Min-Te Sun, Wei-Shinn Ku, An RFID zero-knowledge authentication protocol based on quadratic residues, 9, (14) 2021, pp. 12813–12824.
- [19] Benfano Soewito, Yonathan Marcellinus, IoT security system with modified zero knowledge proof algorithm for authentication, *Egypt. Inform. J.* 22 (3) (2021) 269–276.
- [20] Nitish Andola, Raghav, Vijay Kumar Yadav, S. Venkatesan, Shekhar Verma, Spy-Chain: A lightweight blockchain for authentication and anonymous authorization in IoD, *Wirel. Pers. Commun.* 119 (1) (2021) 343–362.
- [21] Ashutosh Dhar Dwivedi, Rajani Singh, Uttam Ghosh, Raghava Rao Mukkamala, Amr Tolba, Omar Said, Privacy preserving authentication system based on non-interactive zero knowledge proof suitable for internet of things, *J. Ambient Intell. Humaniz. Comput.* 13 (1) (2021) 4639–4649.
- [22] Qi Feng, Debiao He, Huaqun Wang, Lu Zhou, Kim-Kwang Raymond Choo, Lightweight collaborative authentication with key protection for smart electronic health record system, *IEEE Sens. J.* 20 (4) (2019) 2181–2196.
- [23] Zhengwei Ren, Xianye Zha, Kai Zhang, Jing Liu, Heng Zhao, Lightweight protection of user identity privacy based on zero-knowledge proof, in: 2019 IEEE International Conference on Systems, Man and Cybernetics, (SMC), IEEE, 2019, pp. 2549–2554.
- [24] Mu Han, Zhikun Yin, Pengzhou Cheng, Xing Zhang, Shidian Ma, Zero-knowledge identity authentication for internet of vehicles: Improvement and application, *Plos one* 15 (9) (2020) e0239043.
- [25] Ning Xi, Weihui Li, Lv Jing, Jianfeng Ma, ZAMA: A zkp-based anonymous mutual authentication scheme for the iov, *IEEE Internet Things J.* 9 (22) (2022) 22903–22913.
- [26] Antonio Emerson Barros Tomaz, Jose Claudio Do Nascimento, Abdelhakim Senhaji Hafid, Jose Neuman De Souza, Preserving privacy in mobile health systems using non-interactive zero-knowledge proof and blockchain, *IEEE Access* 8 (1) (2020) 204441–204458.
- [27] Nitin Verma, Abhinav Kaushik, Pinki Nayak, A lightweight secure authentication protocol for wireless sensor networks, in: International Conference on Innovative Computing and Communications: Proceedings of ICICC 2020, Springer, 2020, pp. 291–299.
- [28] Usha Jain, Muzzammil Hussain, Jagadeesh Kakarla, Simple, secure, and lightweight mechanism for mutual authentication of nodes in tiny wireless sensor networks, *Int. J. Commun. Syst.* 33 (9) (2020) e4384.
- [29] Pardeep Kumar, Andrei Gurtov, Mangal Sain, Andrew Martin, Phuong Ha, Lightweight authentication and key agreement for smart metering in smart energy networks, *IEEE Trans. Smart Grid* 10 (4) (2019) 4349–4359.
- [30] Wenzheng Liu, Xiaofeng Wang, Wei Peng, Secure remote multi-factor authentication scheme based on chaotic map zero-knowledge proof for crowdsourcing internet of things, *IEEE Access* 8 (2020) 8754–8767.
- [31] Wided Boubakri, Walid Abdallah, Nouredine Boudriga, ZAO-AKA: A zero knowledge proof chaotic authentication and key agreement scheme for securing smart city cyber physical system, *Wirel. Netw.* 27 (6) (2021) 4199–4215.

- [32] Liping Zhang, Yue Zhu, Wei Ren, Yinghan Wang, Kim-Kwang Raymond Choo and Xiong, N. Neal, An energy-efficient authentication scheme based on Chebyshev chaotic map for smart grid environments, *IEEE Internet Things J.* 8 (23) (2021) 17120–17130.
- [33] Vadim Lyubashevsky, Daniele Micciancio, Generalized compact knapsacks are collision resistant, in: *Proc of Automata, Languages and Programming*, Springer, 2006, pp. 144–155.



Zhangquan Wang is currently a professor in Zhejiang Shuren University. His research interests include Intelligent perception and intelligent information processing.



Jiakuan Huang was received the B.S. degree in electronic information engineering from the Zhejiang Shuren University, Hangzhou, China, in 2021. He is currently working toward the M.S. degree in computer science and technology from Changzhou University. His research interests include zero-knowledge proof and network security.



Kelei Miao was received the B.S. degree in electrical engineering and automation from Quzhou University, Quzhou, China, in 2016. Then he received his Ph.D. degree from Zhejiang University of Technology, Hangzhou, China, in 2022. His current research interests include networked control systems and industrial control system security.



Xiaowen Lv was born in 1987. Ph.D, lecturer. Her main research interests include network security and data analysis.



Yourong Chen was received the B.S. degree in communication engineering from Zhejiang University of Technology, Hangzhou, China, in 2004, the M.S. degree in communication engineering communication and information system from Zhejiang University of Technology, Hangzhou, China, in 2007, and the Ph.D. degree in control theory and control engineering from Zhejiang University of Technology, Hangzhou, China, in 2012. Since 2007, he has been teaching in the college of information science, Zhejiang Shuren University. And he is currently a professor in Zhejiang Shuren University. His research interests include network security and internet of things engineering.



Bing Su was received the B.S. degree in mechanical technology and equipment from Nanjing University of Aeronautics and Astronautics in 1994, the M.S. degree mechanical manufacturing and automation from Nanjing University of Aeronautics and Astronautics in 1997, and the Ph.D. degree in mechanical manufacturing and automation from Nanjing University of Aeronautics and Astronautics in 2001. Since 2007, he has been teaching in the School of Computer and Artificial Intelligence, Changzhou University. His research interests include computer applications and wireless sensor networks.



Liyuan Liu is presently serving as an Assistant Professor at Saint Joseph's University, USA. She holds a Ph.D. in Analytics and Data Science from Kennesaw State University, a Master's degree in Analytics from Georgia State University, and a Master's degree in Financial Economics from the University of Detroit Mercy. Her research focuses on various areas such as artificial intelligence in business applications, data security and privacy, and financial technology.



Meng Han was received Ph.D. degree in Computer Science & MBA from Georgia Institute of Technology & Georgia State University. He is currently the Director of Research Center for Innovation through Data Intelligence, ZJU-BJ 100-Young Professor at Binjiang Institute of Zhejiang University. His research interests include data-driven intelligence, data security & privacy, and financial technology, etc.